THE SEPARATION AXIOMS FOR MORPHISMS OF ALGEBRAIC SPACES ARE STABLE UNDER COMPOSITION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement quantifies over 'all of the separation axioms listed in Definition definition-separated'; the proof is written for a general property P, so it covers every axiom in that list at once.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The list of separation axioms in terms of the diagonal. Definition definition-separated, used at proof lines 583-584 and 594-595. Fixes the property P that the diagonal must have.
- The factorization of the diagonal and the property of the comparison arrow. Lemma lemma-fibre-product-after-map, used at proof lines 595-597. Shows the second arrow of the factorization has P as well.
- Stability of representable properties under composition. Section section-representable, used at proof lines 598-600. Gives that the composite of two morphisms with P has P.

Source reference. Original definition 03HL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4571–4571. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03KN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4683–4683. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original section 03HA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4559–4559. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *All of the separation axioms listed in Definition 03HL are stable under composition of morphisms.*

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Let $f : X \to Y$ and $g : Y \to Z$ be morphisms of algebraic spaces to which the axiom in question applies. The diagonal $\Delta_{X/Z}$ is the composition

$$X \longrightarrow X \times_Y X \longrightarrow X \times_Z X.$$

Our separation axiom is defined by requiring the diagonal to have some property $\mathcal{P}$. By Lemma 03KN above we see that the second arrow also has this property. Hence the lemma follows since the composition of (representable) morphisms with property $\mathcal{P}$ also is a morphism with property $\mathcal{P}$, see Section 03HA. $\square$